小兒外科操作手冊

1. History taking

•Children cannot modify their behavior to suit their surroundings and may be frightened and insecure; they must be put at ease.

•The child's and parent's confidence must begained before examination.

•Past history including prenatal and medical history

•Family history and congenital anomalies.

2. Physical examination:

•Gentle and tender--注意隱私權

•When an examination performed, the child will be watching.—注意表情變化

•Peritoneal sign– examination from less pain area

•旁敲側擊，注意小節。

3. Examinations:內容包括

•Non-invasive, fast and cheap study first

•病歷記錄清楚。

.X-ray examinations:

•Removal of patient to X-ray department—difficulties monitor , cold stress and aspiration

•Discussion with a radiologist – may avoid unnecessary examination and will gain more complete information.

•Interpretation by radiologist.

Special X-ray:

•Air – good contrast

•Single bubble sign--pyloric obstruction

•Double bubble sign– Duodenal obstruction

•Three bubble sign--upper jejunal obstrucion

•Pneumatosis intestinalis—NEC

•Foot ball sign: Bowel perforation

• 〝Ground glass〞--meconium peritonitis

4. Antibiotics 的給予:

•Immune compromise condition such as malignancy , premature , liver problem

•High risk procedures such as implantation of prosthetic materials—V-P shunt

•Opening of G-I tract, liver , biliary tract and urinary tract

• The use of indwelling catheters such as nephrostomy tube

•Following splenectomy

5. Fluid requirement:

- Day1 50cc/kg/day
- Day5 100cc/kg/day
- Day10 150cc/kg/day
- Day15 150~180 cc/kg/day
- 3 months ~1 year 100 cc/kg/day
- 1 year 100 cc/kg/day
- 4 year 80cc/kg/day

- 1~10 Kg 100 cc/Kg
- 11~20 Kg 50 cc/Kg
- 20Kg~ 20 cc/kg
- Four quarter scheme
- D2.5S1/2 = D5S +D5W alternately
  shift to D5S1/4 or Taita No 2 later
- Urine output: 1~2 cc/kg/hr

6. Normal values of blood gas

- PH 7.25-7.43
- PCO2 32-45 mmHg
- PO2 > 50 mmHg
- Base deficit/excess -4 to +3
- Standard bicarbonate 18-25 mmol/l

7. Blood transfusion

- Acute blood loss
- Preoperative transfusion
- Iron deficiency anemia
- Haemolytic disease of newborn
- Sickle cell disease
- Other causes of anemia: Thalassemia , hemolysis

Principle:
血色素昇高 1 需要 Whole blood : 6 x Kg cc 或
Pack RBC : 3 x Kg cc
習慣用法:
Whole blood 每公斤給 20 cc，Pack RBC 每公斤給 10 cc，FFP 每公斤給 10~20 cc

8. Colon preparation

- Preoperative antibiotics

•提早 NPO, set IV

•Enterostomy: warm normal saline irrigation,利用合適 Nelaton tube ,洗對 enterostomy，並注意保溫。

•Megacolon: 提早入院，多次 irrigation

9. Content of vomit

•Clear: gastric content – suspect pyloric obstruction , replaced by half saline

•Yellowish: high level small bowel obstruction , replaced by L-R

•Greenish: low level bowel obstruction, replaced by L-R

•Stool: colon obstruction

10. Abdominal distension

•Pyloric or duodenal obstruction: upper abdominal distension , X-ray: few bowel loops– may need elective surgery

•Ileal or colon obstruction: moderate to severe abdominal distension, X-ray: many bowel loops—may need urgent surgery

通則 1：

•NPO : < 3 years old--6hrs , > 3 years old– 8hrs ，G-I tract surgery NPO 要更長。

•NG or OG: Fr 5 for feeding .Fr 8 以上 for decompression

•Foley: if condition stable and I/O balance may remove next day

•所有的 stent 不要任意拔除，如果掉落，往上通報，不可隨意再插入。

通則 2:

•Invasive procedures: isolation

•多對病人及家屬付出關愛，建立良好醫病關係。

●多問、多看及多通報